

Lab 3 Plan

Important Video Links:

Lecture 1: <https://youtube.com/watch?v=rSTYChoXSk>

Lecture 2: <https://youtu.be/kJBImzTXcxU>

Important Slides:

[ALU 1 - Adder & Subtractor v2.pdf](#)

[ALU 5 - Final 4-bit ALU Design v2.pdf](#)

[RF & Memory 2 - Final Register Set Design v1.pdf](#)

20 min	4 bit Adder circuit with top level circuit (Recap)
30 min	4 bit AND circuit with top level circuit (also Tunnelling)
30 min	4 bit LEFT SHIFT circuit with top level circuit
30 min	4 bit ALU supporting ADD, AND & SHL operations
30 min	4 bit Register

Lab 2 Plan

Important Slides:

[uVision Project Tutorial](#) (STM32F103C8 will be selected instead of STM32F100RB)

[ARM Assembly Slide](#)

Reading Book 1 Chapter 6

Software Installation

Keil uVision IDE Software Installation:

Version: 5.39.0.0 (Fill up following form to get the Download Link)

Download Link: <https://www.keil.com/demo/eval/arm.htm>

Video Link: <https://www.youtube.com/watch?v=buQtJjgw2pE>

Before	Install all the softwares using following manual (link) before attending the lab
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Lab	
10 min	Installation Help: Help regarding Logisim Evolution Installations
60 min	Teacher Task: Lab Example 1-7
10 min	Break
70 min	Teacher Task: Lab Example 8-11

Lab 1 Plan

Important Slides:

[Introduction 3 - VLSI & FPGA v2.pdf](#)

[Introduction 2 - Analog Electronics II \(CMOS\) v2.pdf](#)

[ALU 1 - Adder & Subtractor v2.pdf](#)

[HDL 1 - Combinational Circuit.pdf](#)

[HDL 2 - Sequential Circuit.pdf](#)

Before Lab	Install all the softwares using following manual (link) before attending the lab					
10 min	Installation Help: Help regarding Logisim Evolution Installations					
30 min	Teacher Task: Simulation of NOT, NAND and NOR gate using NMOS transistors Students Task (Optional): Simulation of AND/OR Gate using only NMOS transistors					
30 min	Teacher Task: Simulation of 1-bit Adder Circuit using logic gates Simulation of 4-bit Adder Circuit using logic gates Student Task (Optional): Simulation of 2 to 4 Decoder using logic gates. Draw the following in white board to help them.					

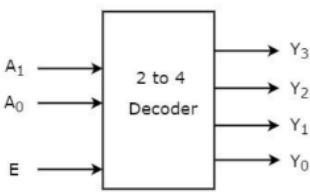


Table: Truth Table of 2 to 4 Decoder

Enable	Inputs		Outputs			
E	A1	A0	Y3	Y2	Y1	Y0
0	x	x	0	0	0	0
1	0	0	0	0	0	1
1	0	1	0	0	1	0
1	1	0	0	1	0	0
1	1	1	1	0	0	0

$$Y_3 = E \cdot A_1 \cdot A_0$$

$$Y_2 = E \cdot A_1 \cdot A_0'$$

$$Y_1 = E \cdot A_1' \cdot A_0$$

$$Y_0 = E \cdot A_1' \cdot A_0'$$

	$Y_3 = E . A_1 . A_0$ $Y_2 = E . A_1 . A_0'$ $Y_1 = E . A_1' . A_0$ $Y_0 = E . A_1' . A_0'$
15 min	Break and Installation Help: Help regarding Verilog Installations
30 min	Teacher Task: HDL code of 1-bit Adder Circuit with test bench (using logic gates + using operators) HDL code of 4-bit Adder Circuit with test bench and wave simulation Student Task (Optional): HDL code of 2 to 4 Decoder Circuit with test bench (using logic gates)
30 min	Teacher Task: HDL code of 2 to 4 Decoder Circuit (using ALWAYS block) HDL code of 1-bit FF Circuit with test bench
Home Work	Student Task: Simulation and HDL code of all circuits of HDL 1 and HDL 2 slides.

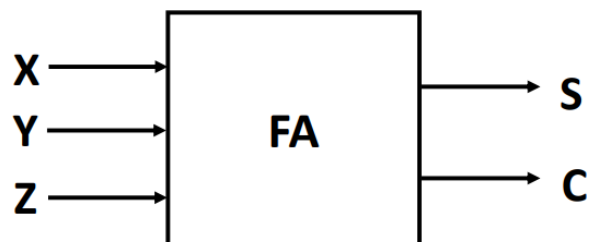
Practice Problems:

1. Logic Circuits using Logisim Evolution:

- a. Design an 4-bit Adder circuit using top level 1-bit Adder circuit with following logic equations:

$$S = X \oplus Y \oplus Z$$

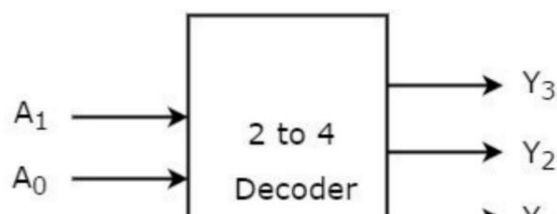
$$C = X Y + (X \oplus Y) Z$$



- b. Design a 2 to 4 Decoder circuit with its Top Level Circuit block (with inputs and outputs pins) with following logic equations:

$$Y_3 = E . A_1 . A_0$$

$$Y_2 = E . A_1 . A_0'$$



- c. Design a logic circuit using NMOS transistors based on following logic equations: $AB+C$.

2. ARM Assembly:

- a. Construct an equivalent ARM Assembly Program of following C Program:

```
//Do not use Register as variable
int a = 10;
int b = 20;
int c = a+b;
```

- b. Construct an equivalent ARM Assembly Program of following C Program:

```
int a = 10; int count = 0;

for(a=0; a<10; a++)
{
    if(a==5) { count=count+1; }
    else if(a >= 7) { count = a++; }
    else { count = a--; }
}
```

3. HDL Code:

- a. Construct 3 to 8 Decoder in Verilog HDL along with a test bench using **if else** statement.
- b. Construct a 4 bit down counter using synchronous design methodology in Verilog HDL along with a test bench.